COMSATS University Islamabad, Lahore Campus

Department of Computer Engineering · BS Computer Engineering — Final Year Project

NeuroMedica: An AI-Powered Medical Imaging & Knowledge Assistant

NeuroMedica

Abstract

NeuroMedica is a single, **explainable AI platform** for medical education that unifies four AI modalities — **imaging, signal, document, and language intelligence** — in one web application. It interprets **chest X-rays** and **ECGs**, digitises handwritten **prescriptions** via OCR, and answers clinical questions through **retrieval-augmented generation** grounded in trusted literature. Every output is transparent, citation-backed, and feeds an editable structured report — built strictly as an **educational decision-support aid, not a clinical substitute**.

Problem & Objectives

Today's medical-learning AI tools are **fragmented, opaque "black boxes,"** and prone to ungrounded output. NeuroMedica integrates them into one trustworthy system:

- AI **Chest X-ray** analysis with Grad-CAM visual explanations.
- AI **ECG** analysis recognising rhythm patterns & beat types.
- AI **OCR** to digitise prescriptions into editable records.
- Reference-grounded **Medical Q&A** with high-precision citations.
- **Symptom-to-disease** exploration with educational write-ups.
- Unify all modules to generate **structured reports**.

Key Outcomes

- ✓ **OCR near-human** accuracy (93.4%) with only 0.2% hallucinated words — safe for medication transcription.
- ✓ **Chest X-ray** DenseNet-121 reaches 0.846 Macro AUC-ROC across 14 pathologies.
- ✓ **ECG** weighted 1D-CNN achieves 0.94 weighted F1 over 17,511 test beats.
- ✓ **RAG** answers stay grounded & traceable; CRAG declines when evidence is insufficient.

Impact & SDGs

- **SDG 3** **Good Health** — trains the next generation of clinicians with safer, evidence-based AI.
- **SDG 4** **Quality Education** — accessible, multimodal, explainable learning anywhere.
- **SDG 9** **Innovation** — integrated, responsible AI infrastructure for healthcare education.

System Architecture & Methodology

Unified Web Workspace — single login · single interface
Next.js 15 frontend · FastAPI backend · Supabase (PostgreSQL) · GPU inference engine

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IMAGING & SIGNAL INTELLIGENCE

Chest X-ray Analysis

X-ray → **DenseNet-121** (CheXNet transfer learning, dual-head) → 14-pathology multi-label + Normal/Abnormal → **Grad-CAM** heat-map

ECG Signal Analysis

ECG image → signal digitiser → **1D-CNN** (class-weighted) → AAMI 5-class beats: N · S · V · F · Q

DOCUMENT & KNOWLEDGE INTELLIGENCE

OCR — Prescription Digitisation

Prescription image → **Qwen2-VL-7B-Instruct** (4-bit nf4) → validated structured JSON → editable patient record

RAG Medical Q&A

Query → hybrid retrieval **(PubMedBERT+Qdrant · BM25 · Neo4j)** → RRF → rerank → **CRAG** self-eval + PubMed fallback → **Claude** → cited answer

Symptom Explorer

Symptoms → retrieval-grounded differential diagnosis → educational treatment notes

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Explainable, Citation-Backed Output
predictions · confidence badges · sources → editable structured clinical report

Datasets & Technology

Chest X-ray — NIH ChestX-ray14	14 pathology classes
ECG — MIT-BIH Arrhythmia (AAMI EC57)	5 beat classes
OCR — IAM Handwriting + prescriptions	VLM digitisation
Knowledge — PubMed / medical literature	RAG corpus

FE: Next.js 15 · **BE:** FastAPI · Python · **DB:** Supabase · **Imaging:** PyTorch · **Signal:** TensorFlow/Keras · **OCR:** Qwen2-VL-7B · **LLM:** Claude · **Vector:** Qdrant · PubMedBERT · **XAI:** Grad-CAM

Safety & Explainability

- **Grad-CAM** heat-maps reveal *why* the X-ray model reached a finding.
- RAG answers carry **inline citations, confidence badges & disclaimers**; the CRAG guardrail refuses ungrounded replies.
- OCR enforces strict JSON — **no hallucinated drug names**. Educational-use-only; no real-time patient data.

Results & Evaluation

Chest X-ray (DenseNet-121 · 14 classes)

Macro AUC-ROC **0.846** Binary AUC **0.787** Binary F1 **0.674**

Best classes	AUC	Hardest classes	AUC
Edema	0.926	Cardiomegaly	0.901
Hernia	0.924	Pneumonia	0.740
Emphysema	0.911	Infiltration	0.713

ECG (Weighted 1D-CNN · MIT-BIH · 17,511 beats)

Beat class	Prec.	Recall	F1	Support
Normal (N)	0.99	0.94	0.96	14,494
Supraventricular (S)	0.47	0.82	0.59	445
PVC / Ventricular (V)	0.81	0.96	0.88	1,158
Fusion (F)	0.28	0.88	0.42	128
Unknown / Paced (Q)	0.98	0.98	0.98	1,286
Weighted Avg	**0.96**	**0.94**	**0.94**	**17,511**

OCR (Qwen2-VL-7B · IAM · 932 words)

Accuracy **93.4%** Precision **0.957** Recall **0.937** F1 **0.947**

Error mix → Substitutions 4.0% · Deletions 2.4% · **Hallucinated insertions just 0.2%** — safe for medication transcription.

RAG Knowledge Assistant

- Hybrid dense + sparse + graph retrieval with RRF & reranking.
- CRAG self-evaluation with live PubMed fallback; declines when evidence is insufficient.
- Retrieval latency **200–400 ms** (< 500 ms target); every answer carries inline citations + confidence badge.

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